

Yuvraj Jha

Bengaluru, KA, India | +91 8016020299 | yuvrajjha1718@gmail.com | linkedin.com/in/yuvrajjha17 | github.com/Yuvraaj | Portfolio

WORK EXPERIENCE

RedHold AI - Diagrams.so | Early Engineer

Remote, IN | Jun 2026 - Present

- Developed and dark-launched a paid public REST API (**26 operations**), a Model Context Protocol (MCP) server (**13 tools**), and Python/TypeScript SDKs for a platform serving **2,000+ users**, validated by **126 passing API tests** and a clean security review.
- Engineered API reliability and security: an **async job pipeline** (HTTP 202 + polling, de-duplication, self-healing) that ended production 504/CORS failures, plus HMAC-hashed scoped keys (invalid keys = **0 DB hits**), idempotent credit metering (**zero double-charges**), rate limiting, and fail-closed Redis.
- Designed the **Well-Architected Score** (0-100, **19 warning types**, 4 severity tiers) with one-click Fix-all and deterministic structural-integrity repair for LLM-driven diagram edits.
- Improved performance and growth: cut LLM cost/latency via **system-prompt caching** (edits drove **~83% of AI spend**) and unblocked **~512 pages** for search indexing (**~1,400 to ~2,200** Google-indexed pages), wiring the GA4 conversion funnel.

North Eastern Space Applications Centre (NESAC), Dept. of Space | Software Engineering Intern - Geospatial Systems

Meghalaya, IN | May 2025 - Dec 2025

- Spearheaded geospatial engineering for a census-independent population-estimation initiative, powering data-driven research workflows used by **12+ scientists across 8 research projects**, now deployed across **4 space-application centres** in India.
- Architected scalable ETL pipelines for GeoTIFF tiling, preprocessing, and DSM/DTM elevation integration, **improving end-to-end processing speed by 75%** on **45 GB+** of satellite raster data.
- Collaborated with remote-sensing experts to translate domain requirements into **production-grade geospatial software**, implementing spatial data modeling, coordinate reprojection, and geospatial-standards-compliant workflows.
- Enhanced system reliability through **thread-safe parallel processing** and automated data-validation pipelines, standardizing geospatial analytics outputs across large-scale research datasets.

DigitalFortress Private Limited | Software Engineer - AI Platforms

Amaravati, IN | Sep 2023 - Apr 2024

- Implemented an OCR pipeline (Python, OpenCV, Tesseract): **25-30% accuracy gain**, **~40% less manual review**, **~90% precision**.

PROJECTS

Self-Pruning Neural Network - Model Compression Research (PyTorch) [\[GitHub\]](#)

- Designed a neural network that prunes itself during training via learnable per-weight sigmoid gates and an L1 sparsity penalty, learning which connections to remove instead of pruning post-hoc.
- Pruned 92.95% of 7.6M parameters** at under 0.5% accuracy loss on CIFAR-10; a 3-phase lambda warm-up schedule held accuracy **variance to 0.44%** across 34-93% sparsity.
- Packaged as an installable Python library with unit tests, CI, and a research-style report - a reproducible study relevant to model efficiency for on-device and edge inference.

Census-Independent Population Estimation - Geospatial AI (YOLOv8, Rasterio) [\[GitHub\]](#)

- Constructed an end-to-end geospatial analytics pipeline transforming satellite imagery into coordinate-aware datasets via **Rasterio tiling**, **YOLOv8 building detection**, and OpenStreetMap integration, classifying residential vs non-residential structures for population modeling.
- Formulated a population-estimation model combining building counts, house-size classification, and **DSM-DTM elevation analysis** to infer floors; delivered an interactive GUI for imagery upload, AI analysis, and coordinate-aware CSV export, **cutting manual analysis time 35%**.

CampusQuery - Retrieval-Augmented Generation (RAG) Q&A Assistant (FastAPI, ChromaDB) [\[GitHub\]](#)

- Built a Retrieval-Augmented Generation (RAG) assistant answering campus queries over **100+ documents** using local MiniLM embeddings and ChromaDB top-k vector search, with a web-search fallback for out-of-corpus questions.
- Added real-time document parsing and a highlight-to-query feature, chaining embeddings, retrieval, and generation through a FastAPI LLM pipeline with a top-k reranking step for higher answer relevance.

Secure Distributed Voting Platform (C++, OpenSSL, AWS) [\[GitHub\]](#)

- Constructed a multithreaded C++ client-server voting platform over TCP/IP sockets (Winsock2) with SQLite persistence and **thread-safe concurrency** for simultaneous multi-user access.
- Secured it with **AES-256-CBC encryption**, **SHA-256 hashing**, and **role-based access control (RBAC)** via OpenSSL; deployed live on AWS EC2 with hardened Security Groups and real-time admin monitoring.

SKILLS

Programming: Python, C++, Java, TypeScript, JavaScript, SQL

ML and AI: Machine Learning, Deep Learning, PyTorch, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Transformers, Model Compression (Pruning), Natural Language Processing (NLP), Computer Vision (OpenCV, YOLOv8), Scikit-learn, TensorFlow, LangChain, GDAL, Rasterio, NumPy, Pandas

Backend and APIs: FastAPI, REST APIs, Model Context Protocol (MCP), Server-Sent Events (SSE), Spring Boot, Flask, Microservices, SQLAlchemy, Alembic, Redis, Async Job Pipelines, API Security (HMAC, rate limiting, idempotency)

Frontend: Next.js, React, Node.js | **Cloud/DevOps:** AWS, Azure, Docker, CI/CD, Git, Linux, PostgreSQL, SQLite

Core CS: Data Structures and Algorithms, Operating Systems, Computer Networks, DBMS, Distributed Systems, OOP, Agile/Scrum

EDUCATION

Vellore Institute of Technology, AP - Integrated M.Tech (Dual Degree), Computer Science and Engineering | 2022 - 2027 | **CGPA: 9.30**

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) 2025 Certified: **Generative AI Professional** | **Data Science Professional**

ACHIEVEMENTS

- Unstop Top Rank Holder - #580 globally, #1 at VIT-AP** | **Walmart Sparkplug 2026 Finalist** | **Flipkart GRiD Robotics Semi-Finalist (2024)**
- Tata Imagination Challenge National Semi-Finalist (2024, 2025; 300,000+ participants)** | **IndiGo Aspiring Leaders (IAL) 2025 Finalist**